

Vector Data

Objectives:

Understanding of vector data models as used in GIS.

Keywords:

Vector, Point, Polyline, Polygon, Vertex, Geometry, Scale, Data Quality, Symbology, Data Sources



Fig. 1. *Looking over a landscape you can see the main features, such as roads, houses and trees.*

Overview

- **Vector** data provide a way to represent real world **features** within the GIS environment.
- A feature is anything you can see on the landscape.
- ***Imagine you are standing on the top of a hill. Looking down you can see houses, roads, trees, rivers, and so on (see Fig.1).***
- Each one of these things would be a **feature** when we represent them in a GIS Application.
- Vector features have **attributes**, which consist of text or numerical information that **describe** the features.

Vector data

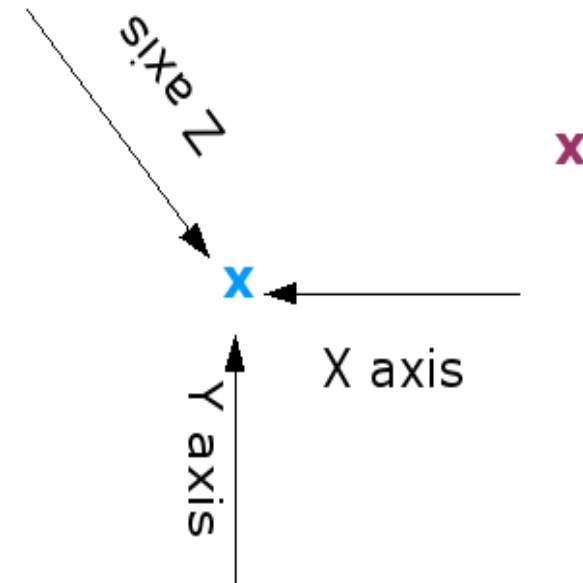
- A vector feature has its shape represented using geometry.
- The geometry is made up of one or more interconnected vertices.
- A vertex describes a position in space using an X, Y and optionally Z axis.
- Geometries which have vertices with a Z axis are often referred to as 2.5D since they describe height or depth at each vertex, but not both.

Vector Point Feature

Fig. 2.

- *A point feature is described by its X, Y and optionally Z coordinate.*
- *The point attributes describe the point e.g. if it is a tree.*

Point Geometry (indicates the x,y and z position of the feature)



Point attributes (describe the feature)

Id, Name, Description

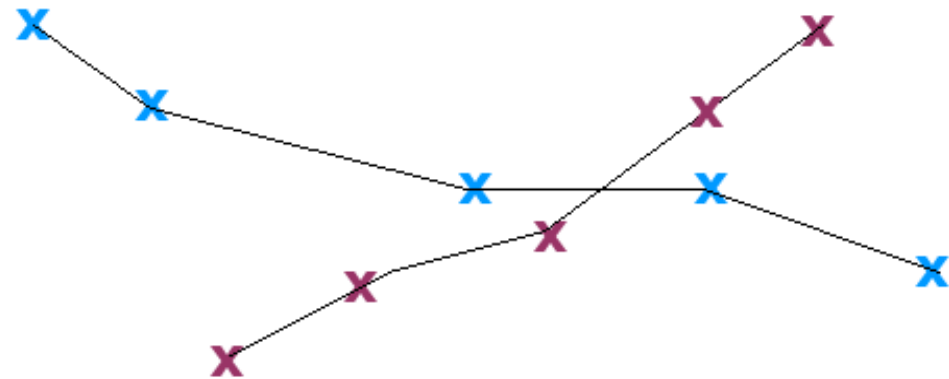
1, Tree, Outside our classroom
2, Light post, At the school entrance

Vector Polyline Feature

Fig. 3.

- A *polyline* is a sequence of joined vertices.
- Each vertex has an *X*, *Y* (and optionally *Z*) coordinate. Attributes describe the *polyline*.

Polyline Geometry (a series of connected vertices that do not form an enclosed shape)



Polyline attributes (describe the feature)

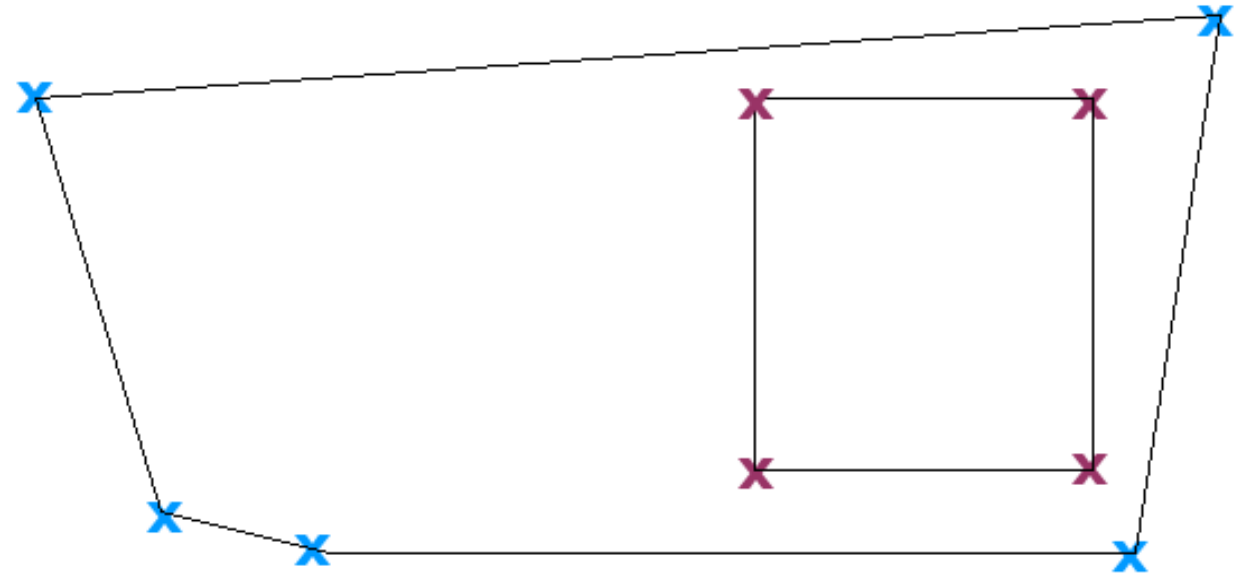
Id, Name, Description

1, Footpath 1, From class to the playground
2, Footpath 2, From the school gate to the hall

Vector Polygon Feature

- Fig. 4.
- *A polygon, like a polyline, is a sequence of vertices.*
- *However in a polygon, the **first and last vertices** are always at the **same position***

Polygon Geometry (a series of connected vertices that do form an enclosed shape)



Polygon attributes (describe the feature)

Id, Name, Description

- 1, School Boundary, Fenceline for the school
- 2, Sports Field, We play soccer here

**Looking back at the picture of a landscape we showed
you further up,**

- you should be able to see the different types of features in the way that a GIS represents them now (see illustration Fig.1.).



Fig. 1. *Looking over a landscape you can see the main features, such as roads, houses and trees.*

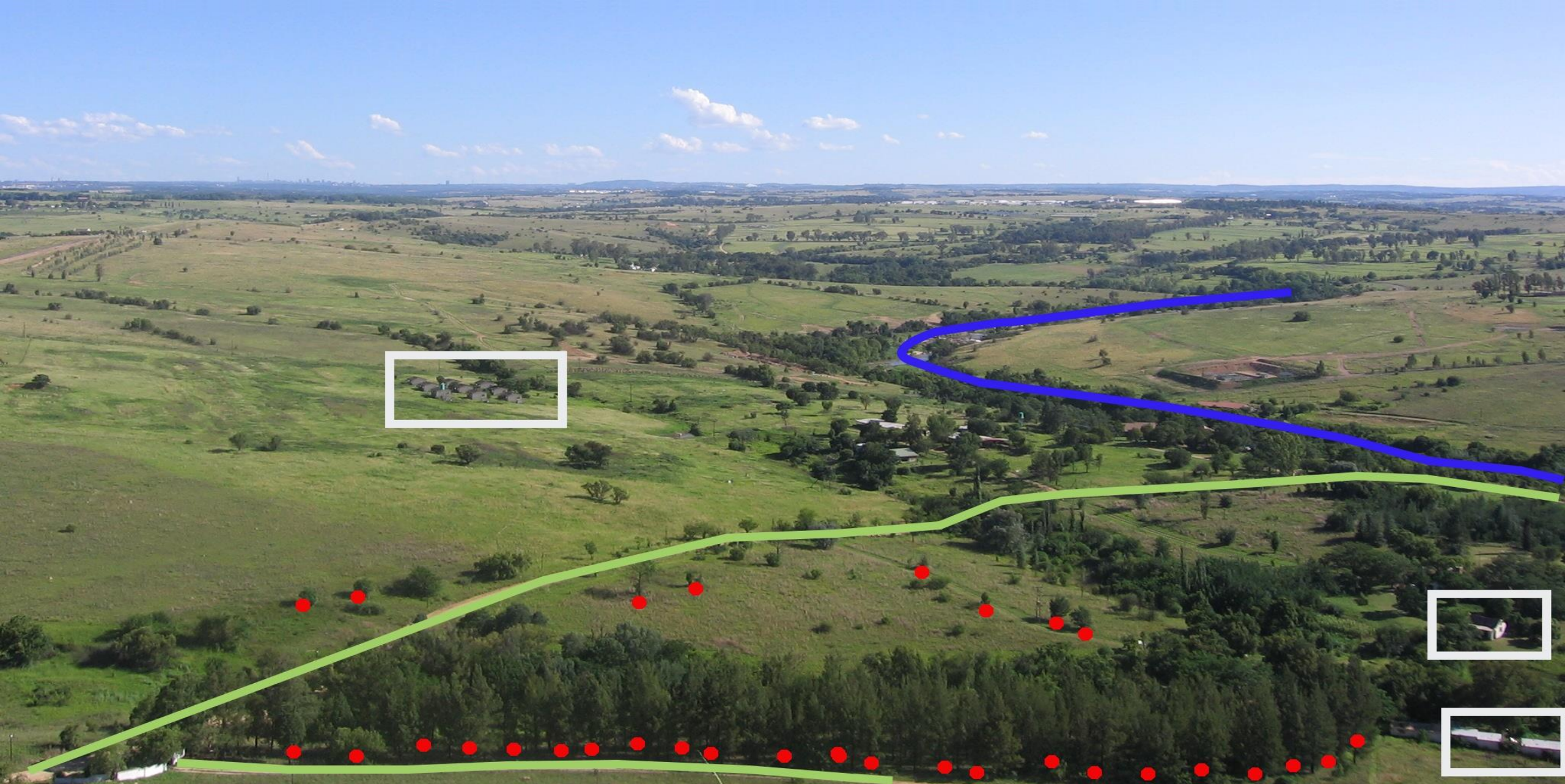


Fig. 5. Landscape features as we would present them in a GIS.

Rivers (blue) and *roads* (green) can be represented as lines, *trees as points* (red) and houses as polygons (white).

Point features in detail

- The first thing we need to realize when talking about point features is that what we describe as a point in GIS is a matter of opinion, and often dependent on scale.

let's look at cities for example.

If you have a small **scale map** (**which covers a large area**), it may make sense to

- represent a city using a point feature.

However as you zoom in to the map, moving towards a larger scale, it makes more sense to

- show the city limits as a polygon.

Polyline features in detail

- Where a point feature is a single vertex, a **polyline has two or more vertices.**
- The polyline is a continuous path drawn through each vertex, as shown in Fig.3.
- When two vertices are joined, a line is created.
- When more than two are joined, they form a 'line of lines', or **polyline.**

Polyline features usage

- A polyline is used to show the geometry of **linear features** such as roads, rivers, contours, footpaths, flight paths and so on.

Sometimes we have special rules for polylines in addition to their basic geometry.

For example

- contour lines may touch (e.g. at a cliff face) but should never cross over each other.
- Similarly, polylines used to store a road network should be connected at intersections.
- In some GIS applications you can set these special rules for a feature type (e.g. roads) and the GIS will ensure that these polylines always comply to these rules.

Polygon features in detail

- Polygon features are **enclosed areas** like dams, islands, country boundaries and so on.
- Like polyline features, polygons are created from a series of vertices that are connected with a continuous line.
- However because a polygon always describes an enclosed area, the first and last vertices should always be at the same place!
- Polygons often have **shared geometry** — boundaries that are in common with a neighboring polygon.
- Many GIS applications have the capability to ensure that the boundaries of neighboring polygons exactly coincide.
- We will explore this in the **Topology** topic later..

- As with **points and polylines**, **polygons** have **attributes**.
- The attributes describe each polygon. For example a dam may have attributes for depth and water quality.

Vector data in layers

- Most GIS applications group vector features into **layers**.
- Features in a layer have the same geometry type (e.g. they will all be points) and the same kinds of attributes (e.g. information about what species a tree is for a trees layer).
- For example
- if you have recorded the positions of all the footpaths in your school, they will usually be stored together on the computer hard disk and shown in the GIS as a single layer.
- This is convenient because it allows you to hide or show all of the features for that layer in your GIS application with a single mouse click.

Editing vector data

- The GIS application will allow you to create and modify the geometry data in a layer — a process called **digitising** — which we will look at more closely in a later Lecture.
- If a layer contains polygons (e.g. farm dams), the GIS application will only allow you to create new polygons in that layer.
- Similarly if you want to change the shape of a feature, the application will only allow you to do it if the changed shape is correct.
- **For example** it won't allow you to edit a line in such a way that it has only one vertex — remember in our discussion of lines above that all **lines must have at least two vertices**.

Creating and editing vector data

- It is an important function of a GIS since it is one of the main ways in which you can create personal data for things you are interested in.
- **For example** you are monitoring pollution in a river.
- You could use the GIS to digitise all outfalls for storm **water drains** (as **point** features).
- You could also digitise the **river** itself (as a **polyline** feature).
- Finally you could take readings of pH levels along the course of the river and digitise the **places where you made these readings** (as a **point layer**).

As well as creating your own data, there is a lot of free vector data that you can obtain and use.

For example, you can obtain vector data that appears on the 1:50 000 map sheets from the Chief Directorate: Surveys and Mapping.

Scale and vector data

- Map **scale** is an important issue to consider when working with vector data in a GIS.
- When data is captured, it is usually digitised from existing maps, or by taking information from surveyor records and global positioning system(**GPS**) devices.

Maps have different scales, so if you import vector data from a map into a GIS environment (for example by digitising paper maps),

- the digital vector data will have the same scale issues as the original map.
- This effect can be seen in illustrations Fig. 6 and Fig. 7. Many issues can arise from making a poor choice of map scale. For example using the vector data in illustration Fig. 6 to plan a wetland conservation area could result in important parts of the wetland being left out of the reserve! On the other hand if you are trying to create a regional map, using data captured at 1:1000 000 might be just fine and will save you a lot of time and effort capturing the data.

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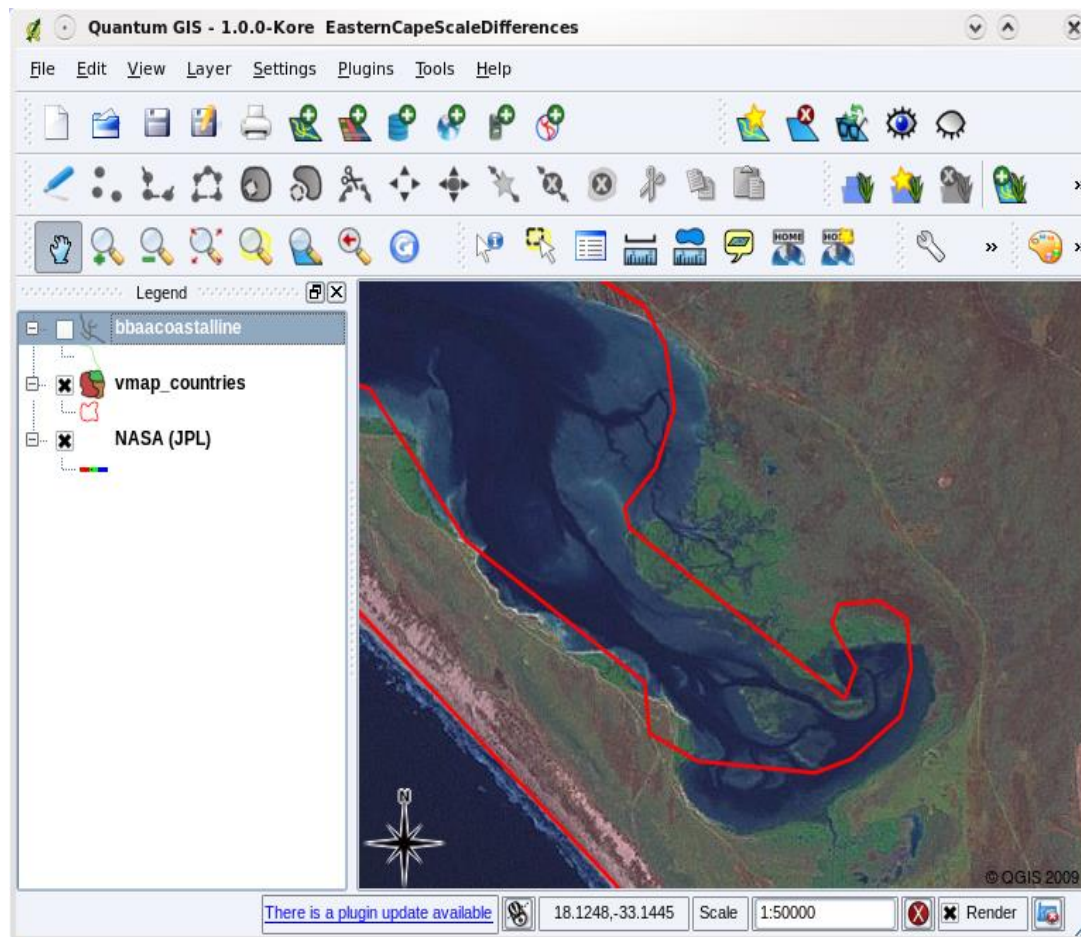


Fig. 6

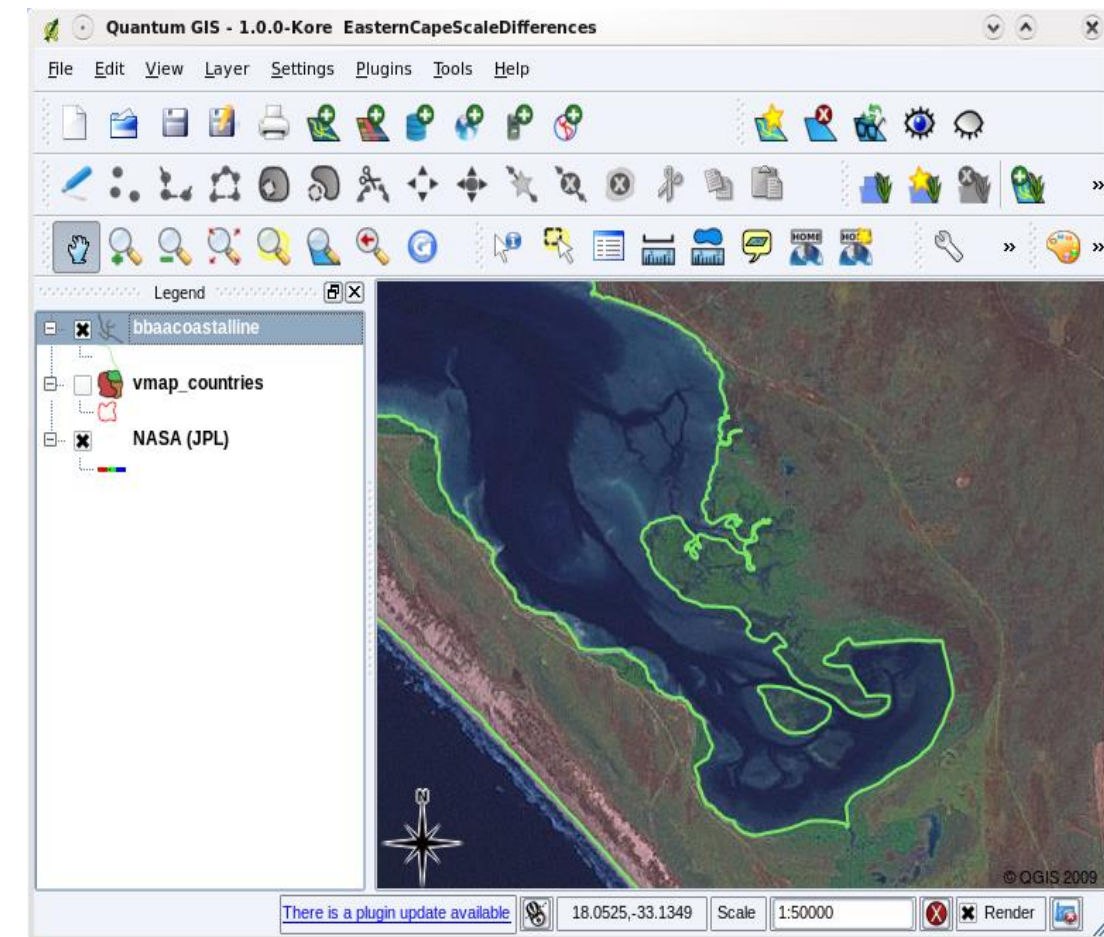


Fig. 7

Symbology

- When you add vector layers to the map view in a GIS application, they will be drawn with random colours and basic symbols.
- One of the great advantages of using a GIS is that you can create personalized maps very easily.
- The GIS program will let you choose colors to suite the feature type (e.g. you can tell it to draw a water bodies vector layer in blue).
- The GIS will also let you adjust the symbol used. So if you have a **trees point** layer, you can show each tree position with a small **picture of a tree**, rather than the basic circle marker that the GIS uses when you first load the layer (see illustrations Fig. 8, Fig. 9 and Fig. 10).

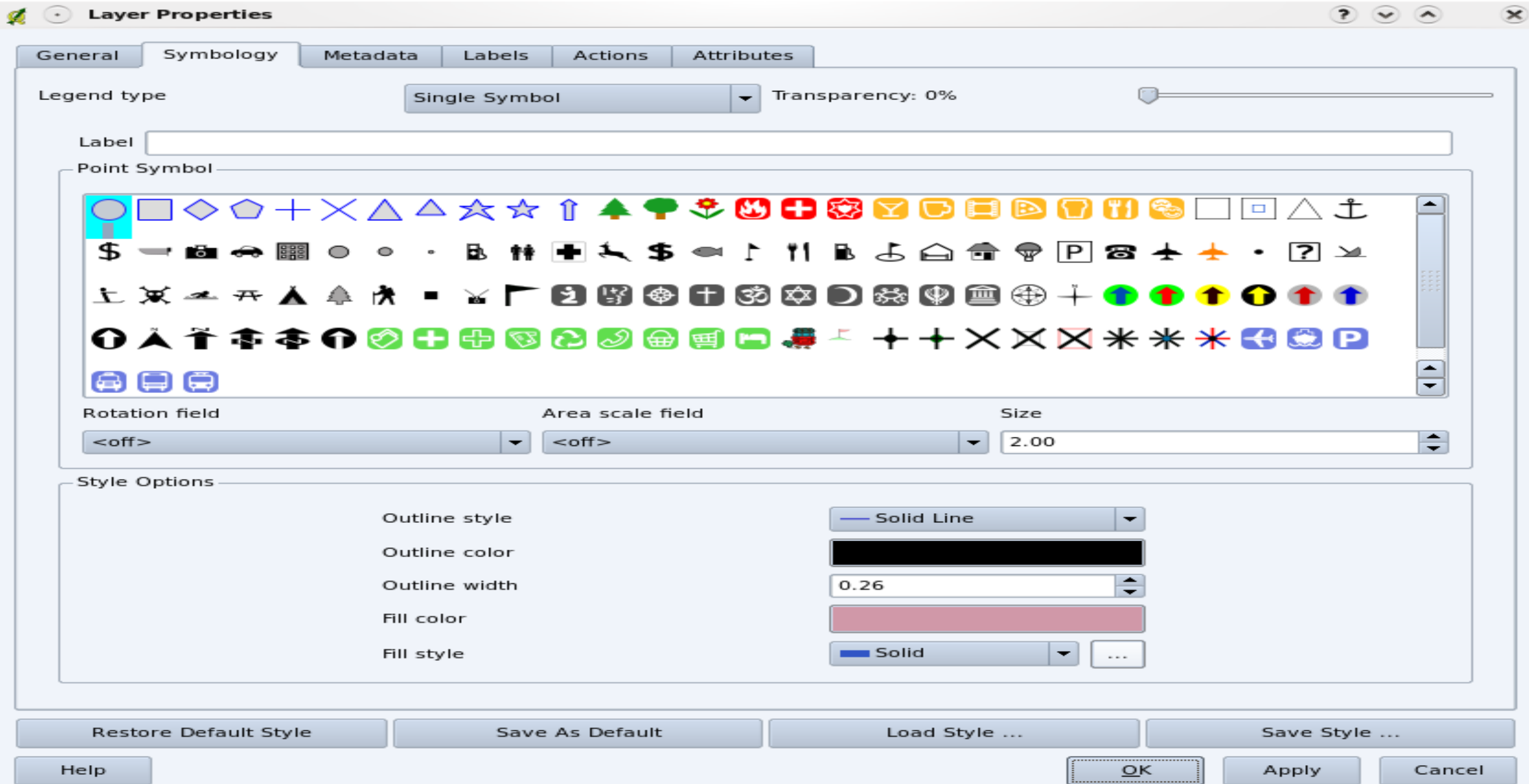


Fig. 8. In the GIS, you can use a panel (like the one above) to adjust how features in your layer should be drawn.

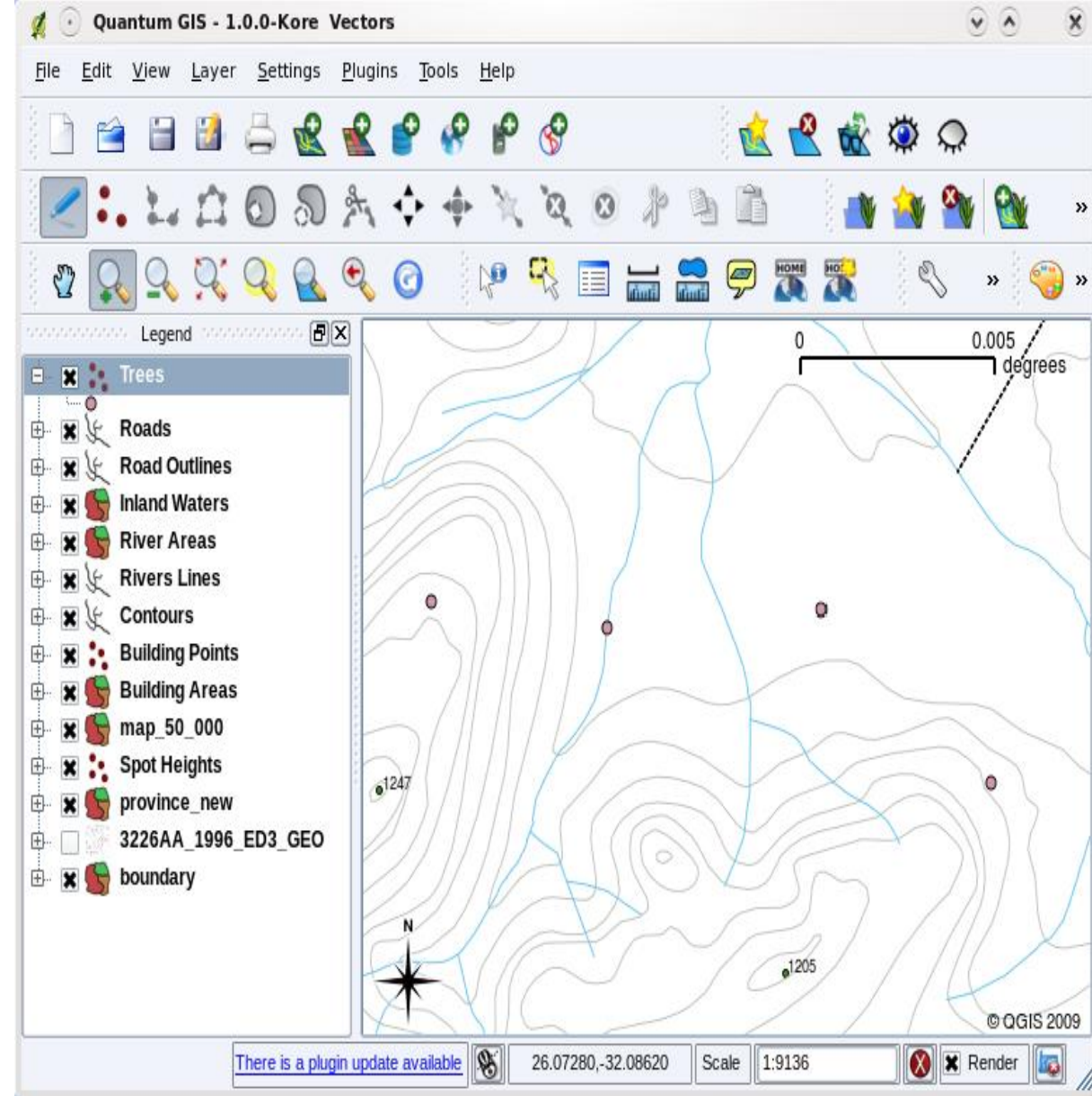


Fig. 9. When a layer (for example the trees layer above) is first loaded, a GIS application will give it a generic symbol.

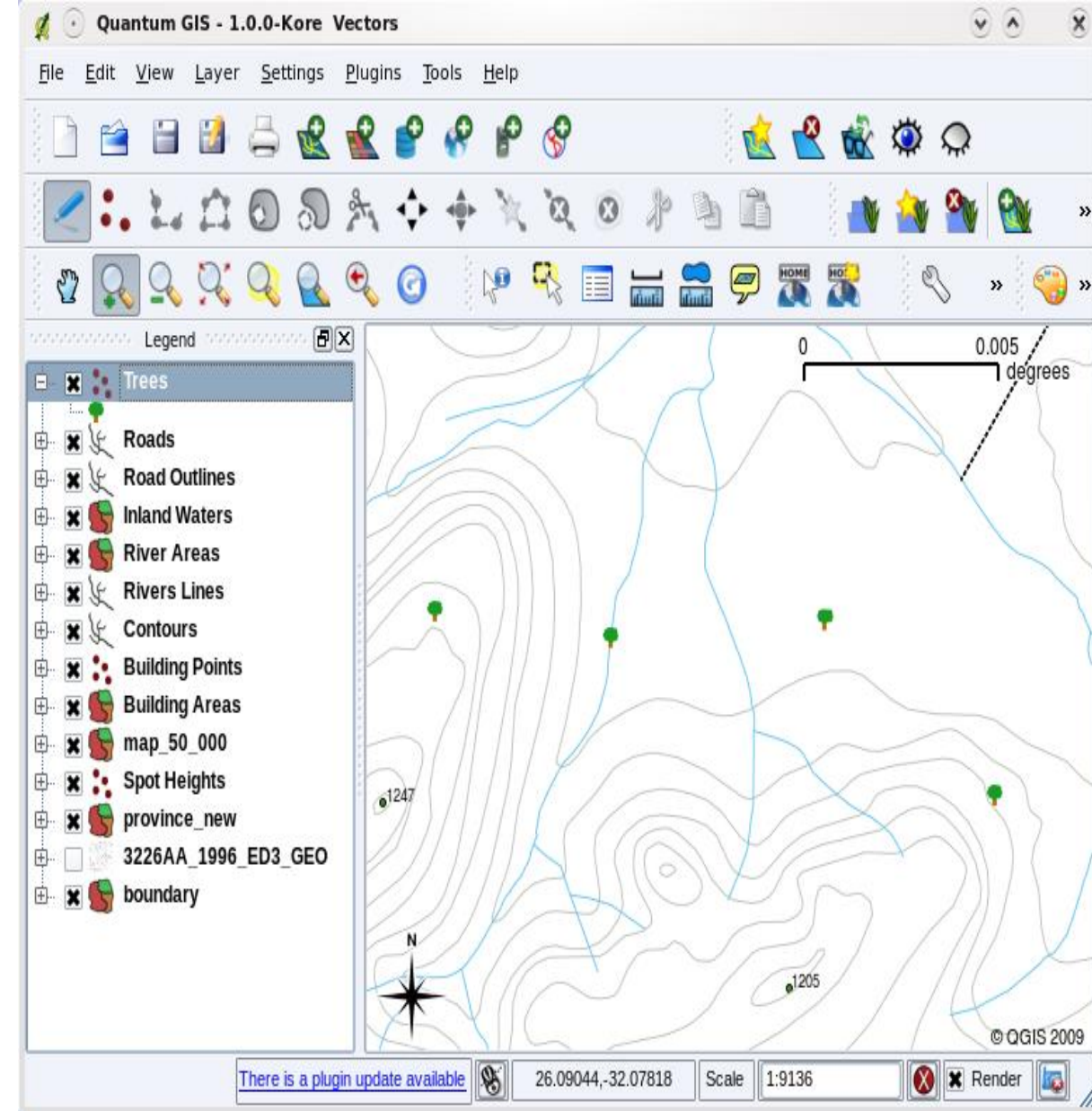


Fig. 10. After making our adjustments it is much easier to see that our points represent trees.

What can we do with vector data in a GIS?

- simplest level we can use vector data in a GIS Application in much the same way you would use a normal topographic map.

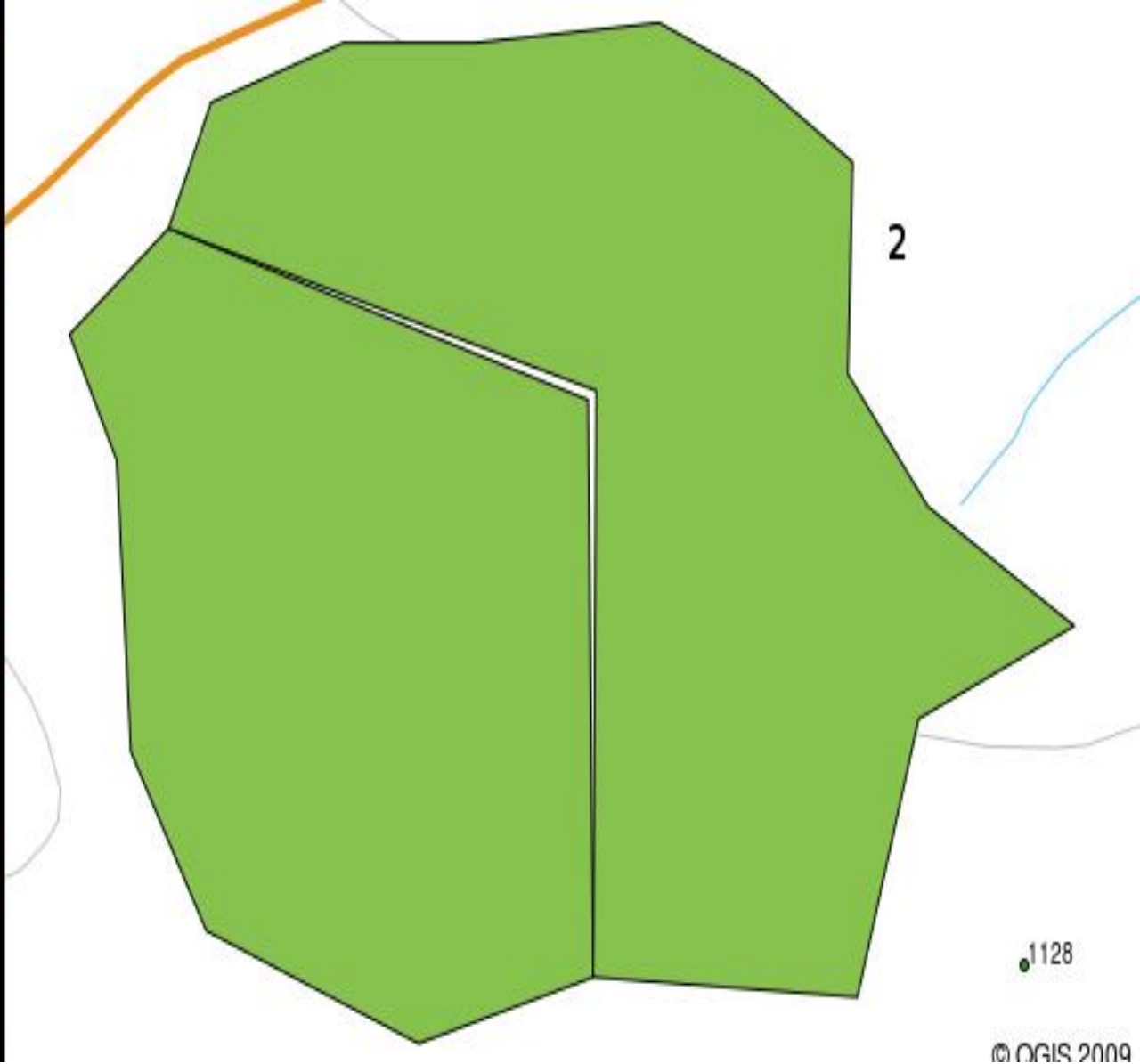
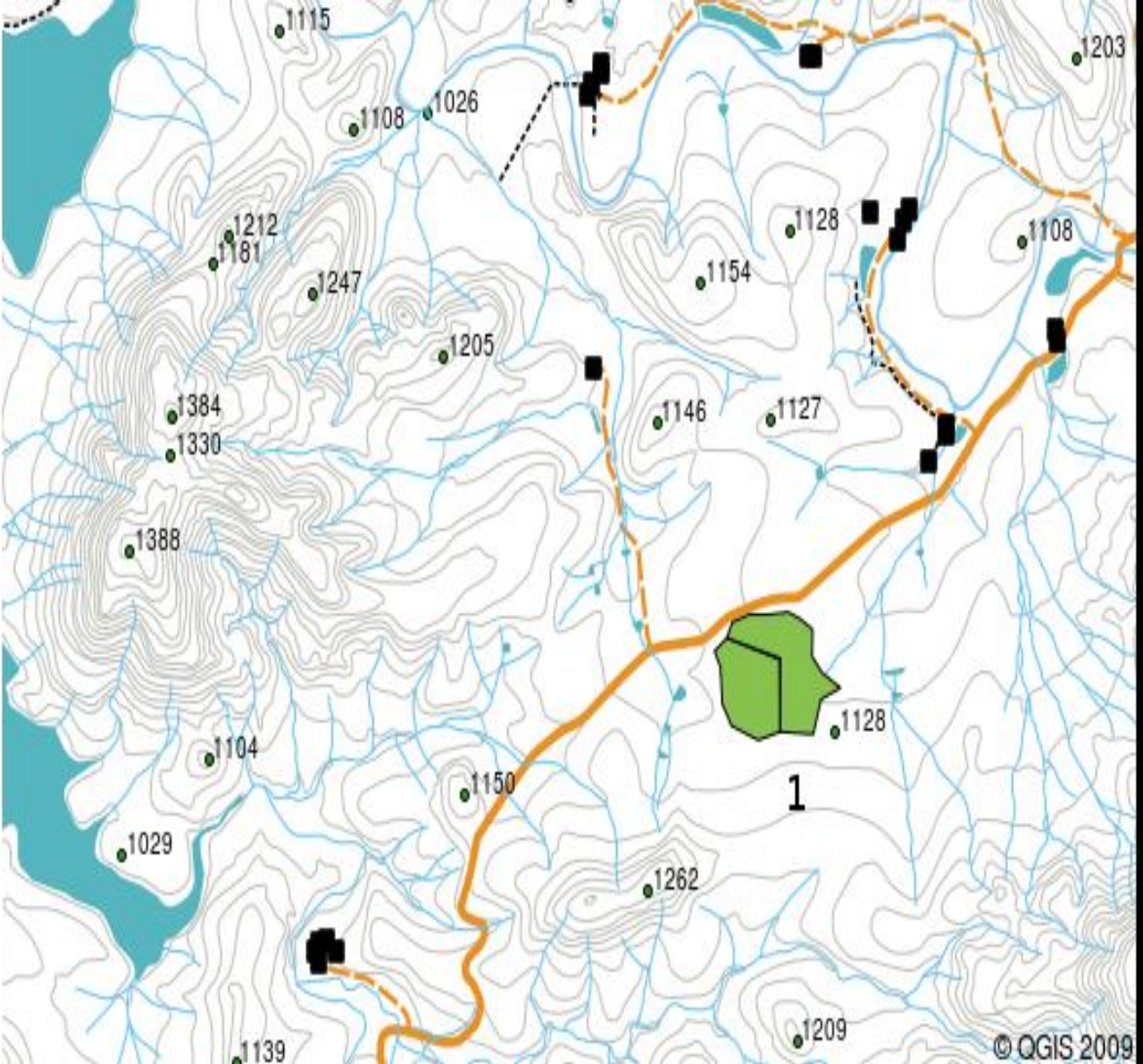
The real power of GIS starts to show itself when you start to ask questions like

- *‘which houses are within the 100m flood level of a river?’;*
- *‘where is the best place to put a hospital so that it is easily accessible to as many people as possible?’;*
- *‘which learners live in a particular suburb?’.*
- A GIS is a great tool for answering these types of questions with the help of vector data.
- Generally we refer to the process of answering these types of questions as **spatial analysis**.

Common problems with vector data

- Working with vector data does have some problems.
- We already mentioned the issues that can arise with vectors captured at different scales.
- Vector data also needs a lot of work and maintenance to ensure that it is accurate and reliable.
- Inaccurate vector data can occur when the instruments used to capture the data are not properly set up, when the people capturing the data aren't being careful, when time or money don't allow for enough detail in the collection process, and so on.
- If you have poor quality vector data, you can often detect this when viewing the data in a GIS.
- **For example slivers** can occur when the edges of two polygon areas don't meet properly (see Fig.11).

- If you have poor quality vector data, you can often detect this when viewing the data in a GIS.
- **For example**
- **slivers** can occur when the edges of two polygon areas don't meet properly (see Fig.11).
- **Overshoots** can occur when a line feature such as a road does not meet another road exactly at an intersection.
- **Undershoots** can occur when a line feature (e.g. a river) does not exactly meet another feature to which it should be connected.
- Fig. 12. demonstrates what **undershoots and overshoots** look like.



- Fig. 11. *Slivers occur when the vertices of two polygons **do not match up on their borders**. At a small scale (e.g. 1 on left) you may not be able to see these errors.*
- *At a large scale they are visible as thin strips between two polygons (2 on right).*



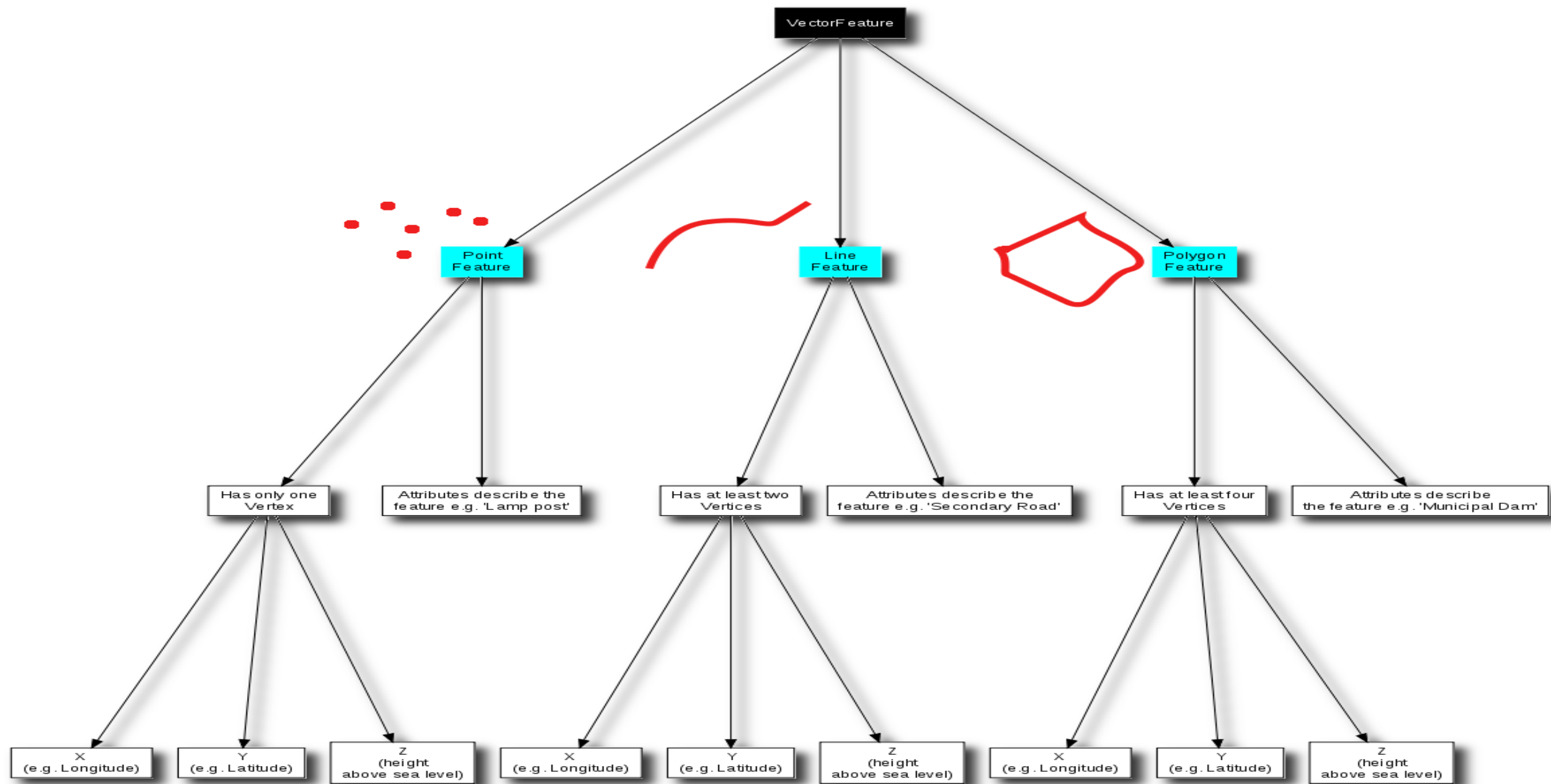
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Fig. 12. Undershoots (1) occur when digitized vector lines that *should connect to each other don't quite touch*. Overshoots (2) happen if a *line ends beyond the line* it should connect to.

What have we learned?

- Let's wrap up what we covered in this Lecture:
- **Vector data** is used to represent real world **features** in a GIS.
- A vector feature can have a **geometry** type of **point**, **line** or a **polygon**.
- Each vector feature has **attribute data** that describes it.
- Feature geometry is described in terms of **vertices**.
- Point geometries are made up of a **single vertex** (X,Y and optionally Z).
- Polyline geometries are made up of **two or more** vertices forming a connected line.
- Polygon geometries are made up of **at least four vertices** forming an enclosed area. The first and last vertices are always in the same place.
- Choosing which geometry type to use depends on scale, convenience and what you want to do with the data in the GIS.
- Most GIS applications do not allow you to mix more than one geometry type in a single layer.
- Digitizing is the process of creating digital vector data by drawing it in a GIS application.
- Vector data can have quality issues such as **undershoots**, **overshoots** and **slivers** which you need to be aware of.
- Vector data can be used for **spatial analysis** in a GIS application, for example to find the nearest hospital to a school.

If we summarized the GIS Vector Data concept in
Figure.....?



Now you try!

- Here are some ideas for you to try :
- Using a copy of a toposheet map for your local area (like the one shown in Fig.13.),
- see can you identify examples of the different types of vector data by highlighting them on the map.
- Think of how you would create vector features in a GIS to represent real world features on your Campus.
- Create a table of different features in and around your Campus and decide whether they would be best represented in the GIS as a point, line or polygon.
- See **Table 1** for an example.

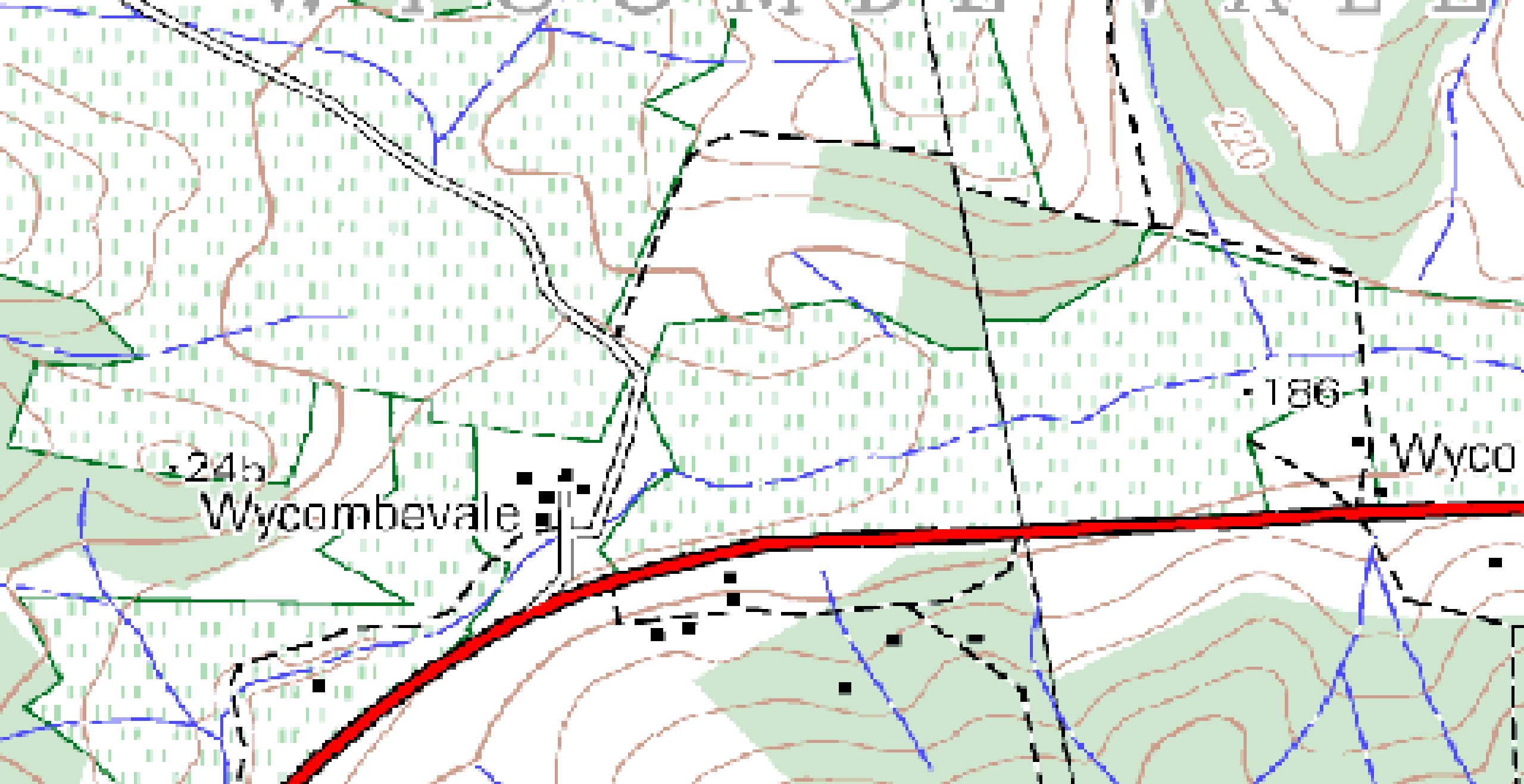


Fig. 12. *Can you identify two point features and one polygon feature on this map?*

Real world feature	Suitable Geometry Type
The school flagpole	
The soccer field	
The footpaths in and around the school	
Places where taps are located	
Etc.	

Table 1:

Create a table like this and decide on suitable geometry types.

- **Something to think about**
- If you don't have a computer available, you can use a toposheet and transparency sheets to show your vector data.
- **Further reading**
- The GIS applications S/w User Guide also have more detailed information on working with vector data .
- **What's next?**
- In the section that follows we will take a closer look at **attribute data** to see how it can be used to describe vector features.